MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8170W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Vertically, base up or down
Weight: 9.5 pounds (4.31 kg) nominal

ABSOLUTE RATINGS: F = 100 MHz

Parameter:	Ef	Eb	Ec2	Ec1	Ib	Pg1
Unit:	Vac	kVdc	kVdc	Vdc	Adc	W
Maximum:						
Class C Teleg:	7.5 ±5%	7.5	1.5	-500	3.0	75
Class C Telep:	7.5 ±5%	5.0	1.0	-500	2.5	75
(Anode mod) Class AB:	7.5 ±5%	7.5	1.5	---	4.0	75
TEST CONDITIONS:	7.5	2.0	0.75	Adj	1.0	---

ABSOLUTE RATINGS: F = 100 MHz

Parameter:	Pg2	Pp	T(Anode core and seal)	Cooling
Unit:	W	kW	°C	Note 1
Maximum:				
Class C Teleg:	250	5.0	250	---
Class C Telep:	250	3.5	250	---
(Anode mod) Class AB:	250	(Note 2) 6.0	250	---
TEST CONDITIONS:	---	---	---	Note 3

GENERAL:

Qualification - Required

Ⓔ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITION:	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMIT MIN	LIMIT MAX	UNIT
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	73	78	Aac
1261	Electrode voltage (grid)	-		0.65	II	Ec1	-95	-127	Vdc
1266	Total grid current	10		0.65	II	Ic1	---	-20	uAdc
1256	Electrode current (screen)	-		0.65	II	Ic2	---	+15	mAdc
1266	Primary grid emission (control)	-	Ic1 = 825 mAdc; t = 15; Ef = 8.25 Vac; anode and screen-grid floating	0.65	II	Isg1	---	-35	uAdc
1266	Primary grid emission (screen)	-	Ec1 = 0; t = 15; Ic2 = 625 mAdc; Ef = 8.25 Vac; anode floating	0.65	II	Isg2	---	-100	uAdc
1231	Peak emission	-	eb = ec1 = ec2 = 2.5 kv	0.65	II	Is	53	---	a
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance (Grounded cathode)	7		---	---	Cin Cout Cgp	108.0 18.0 ---	122.0 23.0 1.0	pF pF pF
1331	Direct-interelectrode capacitance (Grounded grid)	-		---	---	Cin Cout Cpk	48.0 19.0 ---	58.0 24.0 0.16	pF pF pF
1372	Current division (Method B, short pulse)	-	Eb = Ec2 = 1,500 Vdc; Ec1 = -600 Vdc; egk/ib = 11a; tp = 2 us (min); prr = 30 (min)	---	---	egk ic2	--- ---	0 1.25	v a
2214	RF useful output power	3,8	F = 25 to 30 MHz; Eb = 6.0 to 6.6 kVdc; Ib = 2.0 Adc (max) Air: SK-300: 165 cfm(max) SK-300A: 190 cfm (max)	---	---	Ic1 Ic2 Po	--- --- 7.0	80 375 ---	mAdc mAdc kW (Useful)
	Quality conformance inspection, part 3								
---	Service-life guarantee	9		---	---	---	---	---	---
1032	Vibration, mechanical	4,5	Accel = 5 G peak (min); 14 to 200 Hz; accel = 2 G peak (min); 200 to 500 Hz; F = 14 to 500 to 14 Hz; no voltages applied	---	---	---	---	---	---
---	Vibration, mechanical end points:								
1301	Filament current	4,6		---	---	ΔIf	---	1.5	Aac
1261	Electrode voltage (grid)	-		---	---	Ec1	-95	-127	Vdc
1266	Total grid current	-		---	---	Ic1	---	-25	uAdc
1042	Shock, specified pulse	4	Condition A	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Shock, specified pulse end points:								
1301	Filament current	4,6		---	---	Δif	---	1.5	Aac
1261	Electrode voltage (grid)	-		---	---	Ec1	-95	-127	Vdc
1266	Total grid current	-		---	---	Ic1	---	-25	μAd

NOTES:

1. Minimum airflow requirements for incoming air at 50°C maximum at sea level are shown in table I. In all cases of operation a socket which provides forced-air cooling of the base shall be used (such as EIMAC SK-300 with SK-306 chimney, or equivalents, or SK-300A chimney, or equivalents) and maximum anode core and seal temperature ratings shall not be exceeded. The static pressure drop values shown in table I are for the tube and socket, with an SK-306 chimney, or equivalent installed. This pressure drop varies with the amount of escaping air and with the shape and construction of the air director (chimney). Ratings apply for bias voltage less than 500 volts and frequencies less than 100 MHz. Air cooling of the tube shall be increased with increasing negative bias or with increasing frequency. The airflow shall be applied before or simultaneously with electrode voltages, and shall be maintained for a minimum of 2 minutes after all voltages are removed.

TABLE I. Minimum airflow requirements.

Anode dissipation (watts)	SK-300 (See note 1)		SK-300A (See note 1)	
	Airflow (cfm)	Approx press drop (in. H_2O)	Airflow (cfm)	Approx press drop (in. H_2O)
2,000	75	0.4	75	0.4
3,000	100	0.7	105	0.7
4,000	135	1.2	145	1.1
5,000	165	1.8	190	1.5
6,000	200	2.5	230	2.0

2. Applies to carrier only conditions.

3. In all electrical tests involving application of filament voltage, the use of an EIMAC air-system socket, type SK-300 or SK-300A, or equivalent, and air chimney SK-306, or equivalents, will be permissible. Unless otherwise specified under the listed test conditions, forced-air cooling (in a base-to-anode direction) is permitted at a rate of 50 cfm (maximum at sea level, with air at 20°C) for the base and anode; a separate source may be used for the base and anode but neither shall exceed 50 cfm.

(E) 4. This test shall be performed yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

5. Each tube under test shall be subjected to one sweep cycle in each of the three axes X, Y, and Z. One sweep cycle (14 to 500 and back to 14 Hz) shall be covered in 6 to 12 minutes.

6. Any change in filament current resulting from the vibration or shock testing (considered individually) shall not exceed the specified limit for Δif.

7. It shall be allowable to measure Cglg2 and Cglk separately, with all unused elements grounded in each case, and consider the sum to be equal to Cin.

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NOTES: -Continued

8. If an oscillator circuit is used, the screen voltage shall be set to Ec2 = 670 Vdc, and a grid-leak resistance (Rg1) in the control-grid circuit is permissible, with the value of Rg1 adjusted to produce the specified average anode current. If an amplifier circuit is used, protective bias or a grid-leak resistance may be used, with a fixed screen-grid voltage of Ec2 = 500 Vdc, and driving power adjusted to produce the specified average anode current.

9. The tube manufacturer warrants the tube for 1 year from date of shipment, or 1,000 hours of filament life, whichever first elapses. This warranty applies only when the tube is operated within the maximum ratings (see "Absolute ratings" of MIL-E-1). A defective tube shall either be replaced, or at the option of the manufacturer, a credit shall be made in the amount of the original purchase price pro-rated on the basis of 1,000 hours of "filament-on" time.

10. This test shall be the first test performed after the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

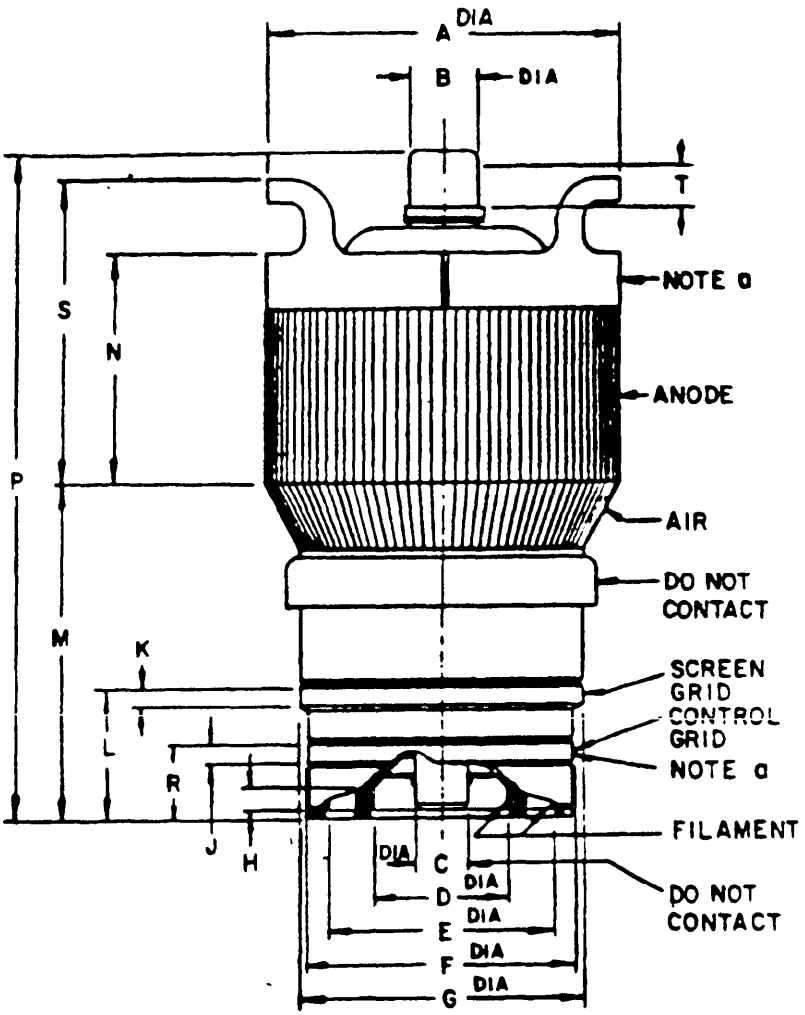
User activities:
Army - MC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
C	.600 (15.24)	.760 (19.30)
D	1.896 (48.16)	1.936 (49.17)
E	3.133 (79.58)	3.173 (80.59)
F	3.792 (96.32)	3.832 (97.33)
G	3.980 (101.09)	4.020 (102.11)
H	.188 (4.78)	
J	.188 (4.78)	
K	.188 (4.78)	
L	1.764 (44.81)	1.826 (46.38)
P	8.625 (219.08)	9.125 (231.78)
R	.986 (25.04)	1.050 (26.67)
T	.375 (9.53)	
Quality conformance inspection, part 3 (see note c)		
A	4.812 (122.22)	4.938 (125.43)
B	.855 (21.72)	.895 (22.73)
M	4.188 (106.38)	4.563 (115.90)
N	2.875 (73.03)	3.250 (82.55)
S	3.875 (98.43)	4.250 (107.95)

NOTES:

a. The total indicator reading (the sum of the positive and negative deflection shown by the indicator when measuring the eccentricity of one surface with respect to another, with the reference axis established) of the screen grid and filament contact surfaces shall not exceed .040 (1.02 mm) with respect to the control grid and anode contact surfaces when the latter surfaces are rotated on rollers at the points indicated by the arrows. (Quality conformance inspection, part 2).

b. Letters H, J, K, N, and T also represents contact surfaces.

(E) c. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8170W.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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